

TIC 60192021

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 9.874251 d
Epoch (t0) 1501.2438 BTJD
Duration 0.1700 d
Depth 10.464 % (104,638 ppm)
SNR 145.2

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 4 (gate ≥ 3)
Robust depth ratio 1.09 (gate ≥ 0.50)
Robust depth sigma 83.8 (gate ≥ 7.0)
Odd/even z-score 0.44 (consistent)
Secondary eclipse detected, phase -0.46, 6.6 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 7 light-curve dip in this sector: 54.7 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 21.12 (gate ≥ 5.0) Compactness 0.87 (gate ≥ 0.50)
Dimming centroid offset 0.16 px (3 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 60192026
0.76 px away, dTmag +1.87 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)

